

Mid Semester Examination 2025 (Set-2)

Name of the Program: B. Tech Petroleum Engineering
Name of the Course: Well logging & Formation Evaluation

Semester: V
Course Code: TPE-502

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
- (ii) Each question carries 10 marks.

Q1	(10 marks)	CO 1
(a)	Explain wireline well logging technology. what are the reservoir parameters measured by logging tools in the well?	
OR		
(b)	How well logging was evolved? What are the tools and techniques adopted in the industry time to time?	
Q2	(10 marks)	CO 2
(a)	Explain sedimentary rock. Write the Five physical properties of the sedimentary rocks.	
OR		
(b)	What is the difference between pore space & porosity? Explain Primary porosity, Secondary porosity, Effective and Total porosity.	
Q3	(10 marks)	CO 2
(a)	What are the three radioactive elements found in the Earth's crust? Draw a graph showing relative abundance in view of Gamma ray emission.	
OR		
(b)	Write formula for volume of shale estimation using GR Log with figure. Calculate Vshale if GR Log= 75, GR Shale= 150 & GR Clean= 40	
Q4	(10 marks)	CO 1 CO 2
(a)	How SP log is generated in bore hole? What is the various type of potential responsible for generating SP log? Explain with figure of ionic movement.	
OR		
(b)	What is the difference resistance & resistivity? Drive a formula for equivalent resistance in series & parallel combinations with figure.	
Q5	(10 marks)	CO 2 CO 3
(a)	What is the principle of caliper tool? What are the uses of caliper log? Write the formula for calculation of mud cake thickness from caliper log.	
OR		
(b)	How GR log will be used for identification of different type of lithologies with depth explain with figure?	

